

Q1 Industrial Sensor Data Processing

START

DECLARE float temperature

DECLARE int storedTemperature

READ temperature

storedTemperature = (int) temperature

PRINT "Temperature Recorded =", temperature

PRINT "Stored Integer value =", storedTemperature

STOP

Eg:

* Input : 36.85

* Output :

* Temperature Recorded = 36.85

* Stored Integer value = 36

Q₂ Production Monitoring system

BEGIN

DECLARE N, i, production, total AS INTEGER

SET total = 0

INPUT N

FOR i = 1 TO N DO

 INPUT production

 total = total + production

END FOR

OUTPUT "Total production =", total

END

* Eg:

Enter N = 5

production values

10

20

30

40

50

o/p:

Total production = 150

Q3 Loan Approval System

Pseudo - code (C++ style)

BEGIN

DECLARE income AS INTEGER

OUTPUT "Enter Annual income :"

INPUT income

IF income $\geq$ 500000 THEN

OUTPUT "Loan Approved - Application is Eligible"

ELSE

OUTPUT "Loan Rejected - Applicant is not Eligible"

END IF

END

* output:

Enter Annual Income : 650000

Loan Approved - Applicant is Eligible

Q4 Employee Payroll System

Pseudo code (C++ style)

BEGIN

DECLARE basicSalary, HRA, bonus, grossSalary AS FLOAT

OUTPUT " Enter Basic salary:"

INPUT basicSalary

OUTPUT " Enter HRA: "

INPUT HRA

OUTPUT "Enter Bonus: "

INPUT bonus

grossSalary = basicSalary + HRA + bonus

OUTPUT " Gross salary = ", grossSalary

END

// output

Enter Basic salary : 30000

Enter HRA : 5000

Enter Bonus : 2000

Gross salary = 37000

Q5

Industrial Machine control

// Pseudo code (c++ style)

BEGIN

DECLARE choice AS INTEGER

OUTPUT "Select Machine Mode"

OUTPUT "1. Start"

OUTPUT "2. Stop"

OUTPUT "3. Maintenance"

OUTPUT "4. Emergency shutdown"

INPUT choice

SWITCH (choice)

CASE 1 :

OUTPUT "machine started"

BREAK

CASE 2 :

OUTPUT "Machine stopped"

BREAK

CASE 3 :

OUTPUT "Machine Under Maintenance"

BREAK

CASE 4 :

OUTPUT "Emergency shutdown Activated"

BREAK

DEFAULT :

OUTPUT "Invalid choice"

END SWITCH

END

// o/p :

Select Machine Mode

1. start

2. stop

3. Maintenance

4. Emergency shutdown

Enter choice : 3

Machine Under Maintenance